Auditing GBFS bike-sharing feeds at country and global scale

A reproducible data-quality taxonomy for open mobility feeds

INPUT

1,509 GBFS systems across 48 countries

transport.data.gouv.fr · MobilityData canonical catalogue

~67,000 French station entries, no semantic audit applied upstream

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FRAMEWORK

Seven-class data-quality taxonomy

five structural errors (A1–A5) + two semantic warnings (A6–A7)

structural errors

- **A1** Out-of-domain inclusion
- **A2** Placeholder capacity
- **A3** Structural over-capacity
- **A4** Geospatial outlier
- **A5** Out-of-perimeter coverage

semantic warnings

- **A6** Zero-capacity dock
- **A7** Null-capacity field

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METHOD

Nine-step idempotent purging pipeline

reversible · logged · unit-tested (n=24, 85% coverage)

kD-tree network geometry, robust 3-sigma statistics, six contextual hybridisations

out-of-sample validation: 12-month MobilityData hold-out (H1, H2 pass)

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RELEASED ARTEFACT

GBFS Audit Catalogue v1.0

DOI 10.5281/zenodo.20125460 · ODbL · Croissant + DCAT-AP + JSON Schema

46,307	46	123	97	30.9%
certified stations	typed columns	audited systems	cities	removed by audit

HEADLINE FINDING

Across the global catalogue, 17 deployments of a single operator publish $c = \text{NaN}$ on every station and would pass any syntactic validator unchanged, yet collectively account for the largest single-operator capacity blackout in the audited corpus (A7 semantic warning, Italy case study).

Syntactic standardisation is necessary but not sufficient.

An open mobility feed is a starting point for research, not a substitute for one.

Fossé & Pallares · CESI LINEACT, Montpellier · 2026 · Submitted to Computer Standards & Interfaces